hydrogen ion concentration. Conversely, decreased ventilation increases CO_2, thus increasing hydrogen ion concentration in the extracellular fluid. Both bicarbonate buffer system and hemoglobin buffer system of erythrocytes are important for the maintenance of pH by lungs.

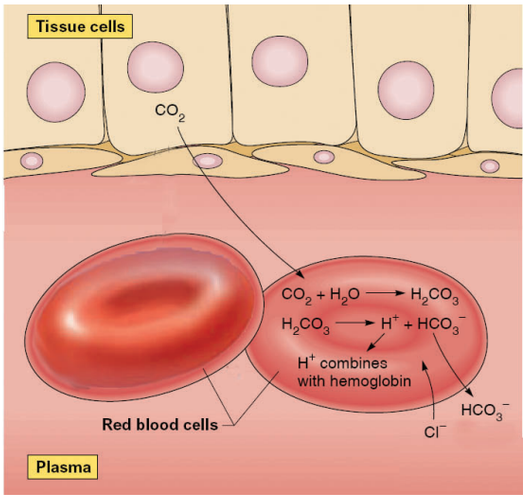


Figure 1.20 – The dissolved carbon dioxide diffuses into erythrocytes from the tissues. It is then converted into carbonic acid by the enzyme carbonic anhydrase. Carbonic acid then dissociates into hydrogen ion and bicarbonate ion. The hydrogen ion then combines with haemoglobin as HHb and the bicarbonate ion diffuses out into plasma. To maintain electrical neutrality, chloride ions diffuse into erythrocytes.



BOHR EFFECT

The unloading of oxygen is increased by the binding of H^+ (released from carbonic acid) to oxyhemoglobin. This is the Bohr effect, and results in increased conversion of oxyhemoglobin to deoxyhemoglobin.



CHLORIDE SHIFT

The exchange of chloride ions for bicarbonate ions as blood passes through the systemic capillaries is called as the chloride shift.

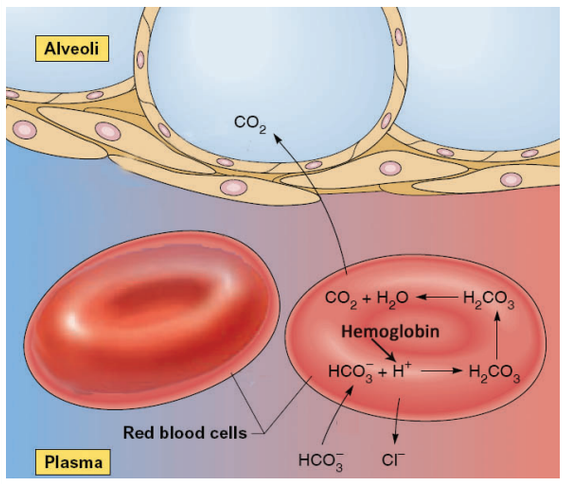


Figure 1.21 – The bicarbonate ions diffuse from plasma, combine with hydrogen ions released from hemoglobin, to form carbonic acid. The carbonic acid is then converted into carbon dioxide and water by the enzyme carbonic anhydrase. The carbon dioxide is then eliminated through exhaled air.

When blood passes through capillaries of systemic circulation, carbonic acid is formed from diffused carbon dioxide and water by the action of the enzyme carbonic anhydrase within the erythrocytes. High carbonic acid concentration favors the dissociation of carbonic acid into bicarbonate and hydrogen ions within erythrocytes. The released H^+ ions bind with deoxy haemoglobin whereas bicarbonate ions diffuse out of erythrocytes into plasma. Thus haemoglobin reduces H^+ ions within erythrocytes in tissues.

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